

Final Project Proposal: Aerodynamic Shape Optimization of Formula One Rear Wing

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1. Problem

In Formula 1, optimizing aerodynamics is the determining factor for the competitiveness of a car in the technical series. Six components dominate the aerodynamic performance of the vehicle, each with their own set of flow regimes, challenges, and functions: the front wing, rear wing, undertray, diffuser, sidepods, and engine cover. Of these, the rear wing alone generates roughly 25–40% of the total vehicle’s downforce and a comparable share of drag, making it critical to optimize [Bhatnagar (2014)].

Balancing the downforce against drag is complicated with varying driving conditions every lap. Under braking and in corners, maximum downforce is desirable for grip and controllability. Typically, "Z-Mode" is activated, rotating the blades of the rear (and front) wing in order to maximize downforce with some extra drag penalty. On straights, maximum efficiency is desirable, so "X-Mode" is activated, turning the wings’ blades to reduce angle of attack, thus reducing drag at the expense of some lost downforce [Barretto (2024)].

The culmination of geometric constraints for the wing, set airfoil shapes, and rotating blades, all under the objective to maximize agility and efficiency on the road for varying driving conditions, makes the F1 rear wing design a rich and multi-objective optimization problem. It is also amenable to simplification: reduced-fidelity aerodynamics and a constrained design space make the problem accessible as a course project. Our project isolates the rear wing and simplifies it to two elements (main and secondary element), each with a center and outboard cross section which can be interpolated between to form a 3D solution. The geometry will be optimized under X-Mode and Z-Mode flap orientations. See Figure 2 for information. Together, these parameters serve as the setup for a multi-stage, multiobjective optimization problem, which is expanded upon in the next section.

In the spirit of the upcoming Grand Prix, and for the love of the sport (and Charles Leclerc), we target the Circuit de Monaco Grand Prix as the course of interest for rear-wing design. Monaco, in addition to being the classic F1 circuit, is a mix of tight, low-speed corners and short straights that result in a high fraction of lap distance being under cornering conditions where the car is operating in the high-downforce regime. This gives the Z-Mode unusual dominance in the design consideration. The makeup of the Monaco course is shown in Figure 1.

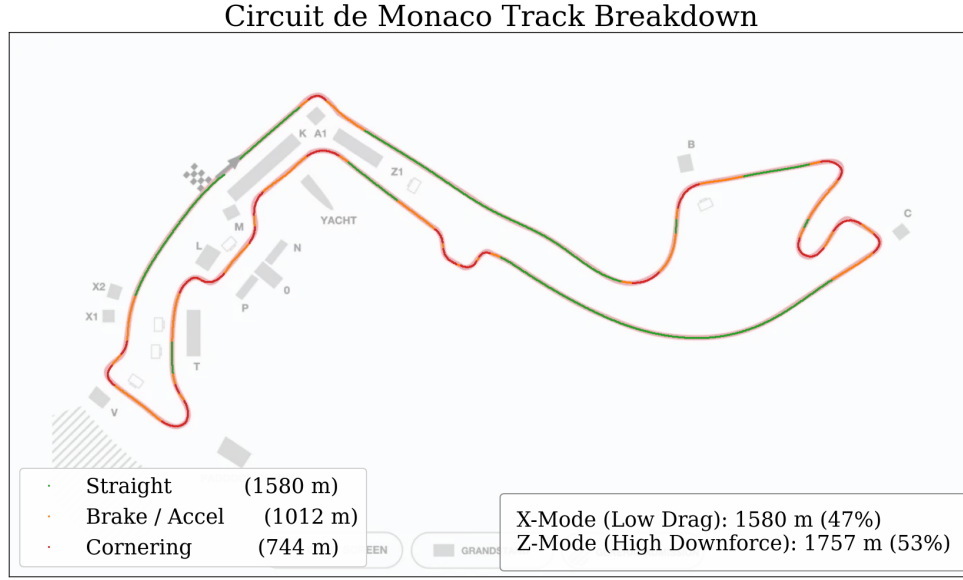


Figure 1: Circuit de Monaco Grand Prix divided into curved, brake/acceleration, and straight portions.

2. Proposed Approach

2.1 General Methodology

Step 0 (Scoring System): Assign scoring system based on the track that we choose (different tracks result in different optimized designs as some tracks have more straights while others have more turns, which affects the degree to which we care about downforce versus drag) and determine what design variables we want to change (angle of attack, thickness, thickness location, max camber, max camber location).

Step 1 (Seed Wing): Analyze the downforce and drag of a seed Formula 1 wing. Note the angle of attack, camber, thickness, camber location, and thickness location so that we can compare how much these design variables change with the optimized design.

Step 2 (First element): Fix the leading edge and chord of the first element to a set value. Optimize the center and outboard sections to maximize points. This will result in two separate, optimized airfoils for the two sections that are experiencing different flow conditions. Design variables: angle of attack, thickness, thickness location, max camber, max camber location. This optimization is done without the second element and once optimized, the shape and design of the first element is set.

Step 3 (Second element): With the optimized first element, now fix the second element leading edge and chord in relation to the first element. Optimize the center and outboard sections to maximize points. Again, this will result in two separate, optimized airfoils for the two sections that are experiencing different flow conditions. Design variables: angle of attack, thickness, thickness location, max camber, max camber location. This optimization is done using the already fixed and optimized first element so that only the second element changes now.

Step 4 (Active Aero): Formula One cars have active aero which allows the second element to change its angle of attack at certain points throughout the track rather than being fixed at one angle the entire race. This is typically done in the straights as the second element reduces to a low angle of attack to minimize drag and maximize speed. We will take the optimized design from above and fix all parameters except for the following: the trailing edge location will be fixed and the angle can change. Rather than the points system, the goal here is to minimize drag. This will tell us what angle the second element should rotate to in the straights.

Step 5 (Comparison): Compare to the seed wing and determine the effectiveness of the new design.

Step 6 (Time permitting CFD Validation): 2D RANS CFD simulations of both sections of both wings. CAD both wings and run 3D RANS CFD simulations.

Assumptions:

- Center section of the wing is no more or less important than the outboard section.
- The incoming flow to the outboard section is freestream flow ($\text{AoA} \approx 0$ degrees and increased velocity) compared to the center section which sees dirty flow due to it being in the wake of the body of the vehicle ($\text{AoA} \approx 5$ degrees and lower velocity).
- The scoring system weighting comes from the makeup of the track itself.

2.2 Pymead and Optimizer Workflow

Python Multi-Element Airfoil Design [Pymead: Lauer (2023) and Lauer (2025)] will be used for geometry generation and aerodynamic analysis, while our own code will handle the optimization. The optimizer selects a set of design variables (angle of attack, thickness, thickness location, maximum camber, and camber location), which are passed into Pymead to generate the airfoil and compute aerodynamic quantities such as C_L and C_D . These values are then used to evaluate our track-based scoring function.

In this setup, Pymead acts only as an analysis tool, while the optimizer determines how the design evolves. The optimizer proposes a design, Pymead evaluates it, and the resulting score is used to decide the next design. This process is repeated until the design converges.

Pymead has a built-in optimization feature that uses the U-NSGA-III genetic algorithm, which will serve as one of the methods in this project. To compare optimization strategies, we will also implement a second algorithm outside of Pymead and apply it to the same workflow. Both methods will use the same design variables, aerodynamic evaluation, and scoring function, allowing for a direct comparison of convergence behavior, computational cost, and final geometry.

2.3 Optimization Algorithm

There are several optimization approaches that are suitable for this problem. Two of the most relevant are NSGA-II (Non-Dominated Sorting Genetic Algorithm II) and Bayesian optimization.

NSGA-II is a population-based genetic algorithm designed for multi-objective optimization problems, making it an ideal candidate for aerodynamic design tasks that involve competing objectives, such as minimizing drag while maximizing downforce. A major advantage

of NSGA-II is that it does not require gradient information. However, this algorithm requires a large number of function evaluations to evolve the population. When coupled with aerodynamic solvers, each evaluation can be computationally expensive, leading to a large overall runtime [Deb et al. (2002), Ullah et al. (2025)].

Bayesian optimization is a surrogate-based method that is ideal for problems where function evaluations are costly. It uses an acquisition function to strategically determine which regions of the design space to explore and exploit. While Bayesian optimization tends to be more efficient, it is generally better suited for single-objective problems [Liu et al. (2022)].

The choice of optimization algorithm will ultimately depend on the final formulation of the problem. Factors such as the number of objectives, constraints, and design variables will play a key role in determining the most appropriate approach.

3. Sketches of Solution (Optional)

In Figure 2 the expected solution for the two airfoil sections is shown. As previously mentioned in the assumptions, the center section sees an incoming flow at a lower velocity and an angle of attack of ≈ 5 degrees, while the outboard section sees an incoming flow at a higher velocity and an angle of attack of ≈ 0 degrees. This can be seen in Figure 2 where the velocity vectors reflect these flow fields. Due to the velocity being slower at the center section, it is expected that both the first and second elements will need to have more camber and be placed at a more aggressive angle of attack in order to generate an adequate amount of downforce. Meanwhile, the outboard section does not need as much camber or an aggressive angle of attack since it will be seeing cleaner air.

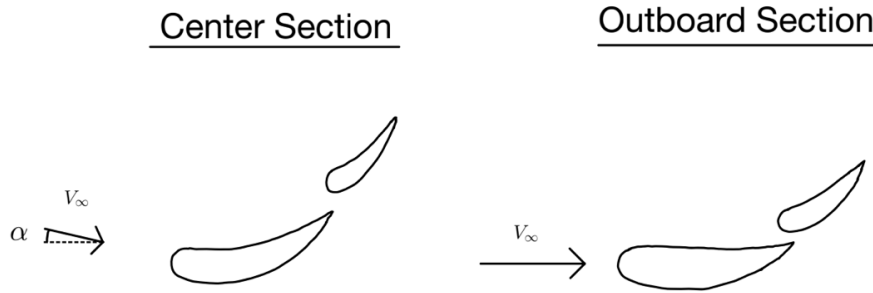


Figure 2: Sketch demonstrating the expected trends in the optimized design for the inboard and outboard sections.

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